

Abstract No.

A conserved astrocyte-like parental cell defines the phylogenetic root of adult-type diffuse glioma

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Adult-type diffuse gliomas are among the most lethal human cancers, yet how genetic clonal evolution gives rise to transcriptional heterogeneity has remained unresolved. We assembled one of the largest single-cell atlases of glioma to date, integrating scRNA-seq, snRNA-seq, and spatial transcriptomics across 69 tumours and over 1.2 million cells spanning IDH-mutant astrocytoma, oligodendroglioma, and IDH-wildtype glioblastoma. Coupling transcriptomes to single-cell DNA and mitochondrial lineage markers, we reconstructed phylogenetic clonal evolution and, by defining the parental clonal population, identified a quiescent, astrocyte-like parental cell (ALPC) at the phylogenetic root of every tumour. Orthogonal DNA- and protein-level assays, in situ spatial localisation, and patient-derived stem-permissive cultures validated the ALPC as a conserved, stem-like, differentiation-competent founder compartment. From this apex we resolved seven recurrent transcriptional states—including neural- and oligodendroglial-lineage and proliferative programs—that recapitulate normal neural stem-cell differentiation trajectories. Despite arising from distinct foundational driver alterations, all glioma types converged on a shared hierarchical architecture in which the quiescent parental compartment gives rise to proliferative and lineage-committed progeny. The IDH axis modulated this common framework rather than redefining it: IDH-mutant tumours were dominated by differentiated lineage-like states with relative immune paucity, whereas glioblastoma showed expansion of ALPC and hypoxia states with increased immune infiltration and mitochondrial oxidative-phosphorylation programs. Together, these findings link clonal evolution to cellular heterogeneity through a conserved developmental framework, repositioning the astrocyte-like parental cell as the lineage-defined root of glioma cell states. Because this founder compartment is shared across subtypes yet sits upstream of all malignant progeny, eliminating it may offer pan-glioma therapeutic opportunities.



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